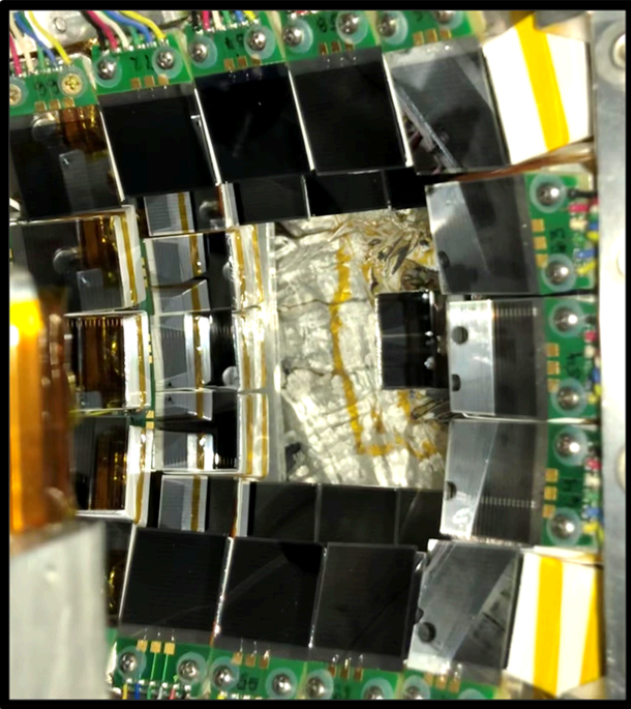


The Texas Automated Particle Identification Routine (TAPIR)

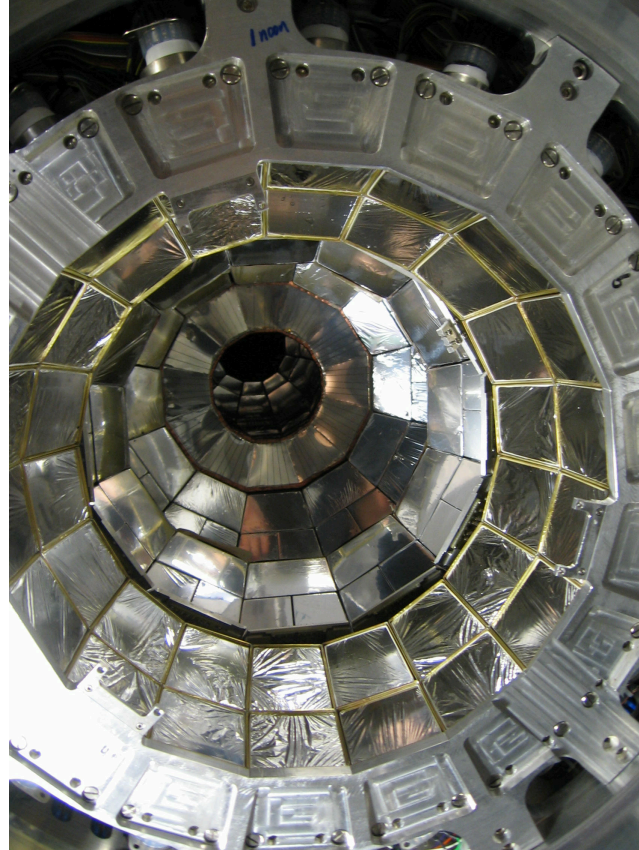
Bryan M Harvey, Spring APS 2026

Telescoped Multidetector Array Experiments

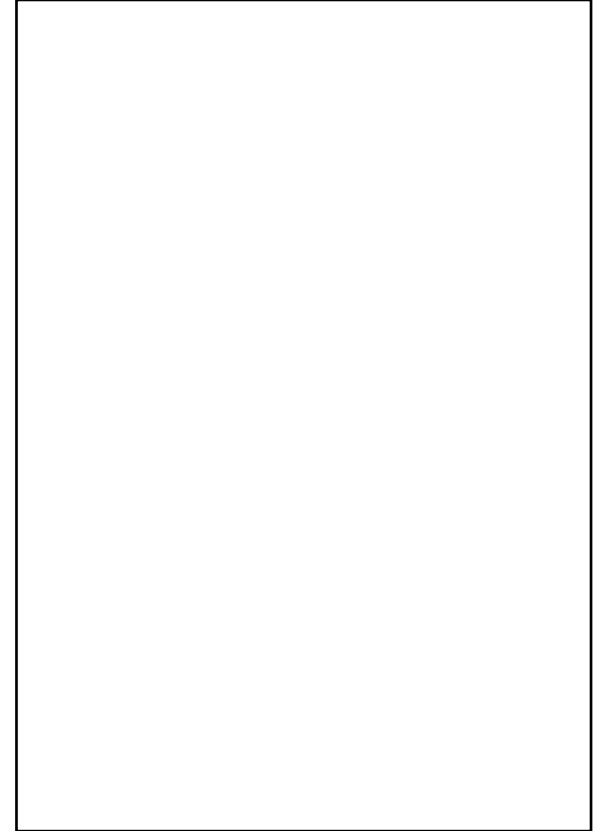
FAUST



NIMROD



FAZIA

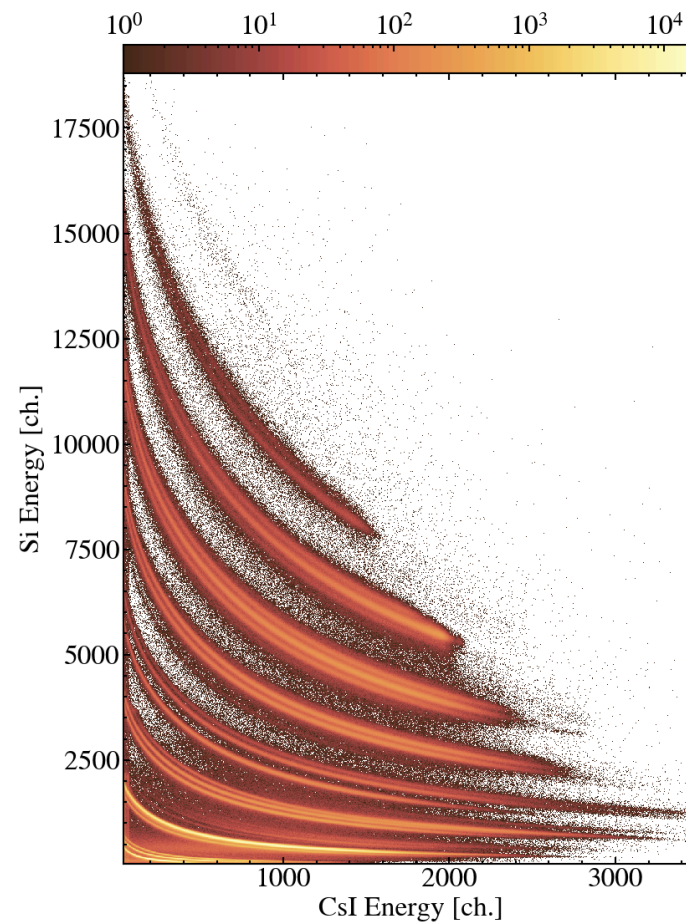


Particle Identification via Energy Loss

FAUST telescope with E and ΔE
labeled

Bethe-Bloche

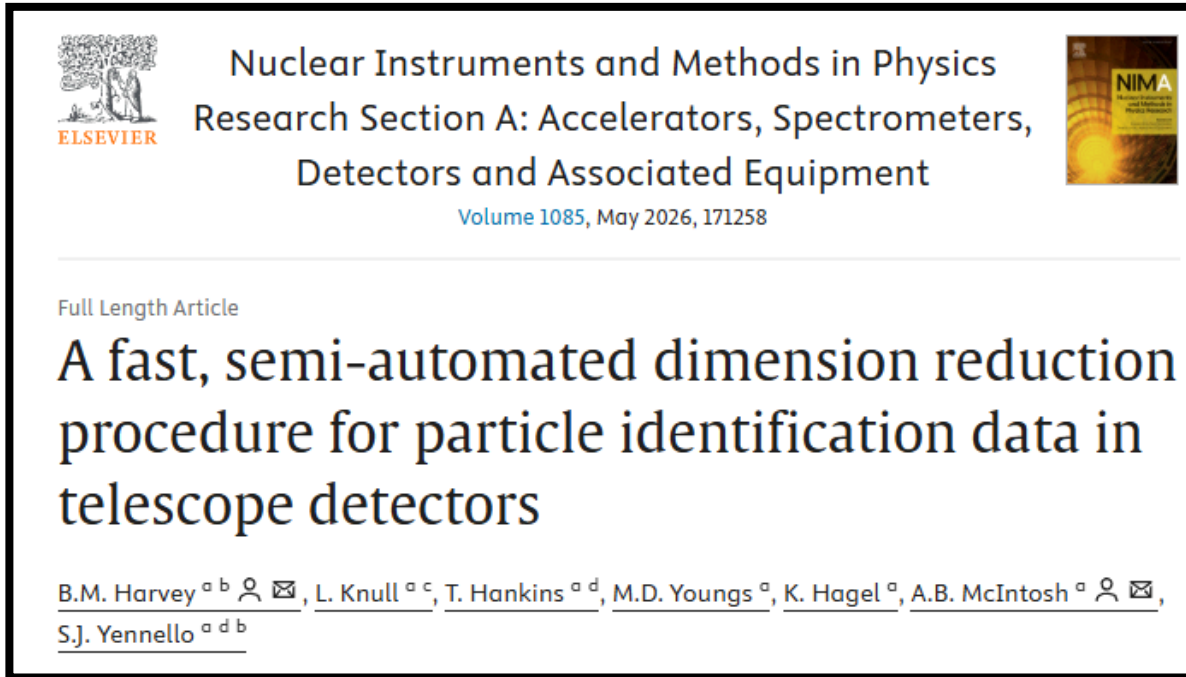
$$-\frac{dE}{dx} \propto \frac{AZ^2}{E}$$



By-Hand Linearizations

ΔE versus E plot with
Travis's by-hand curves

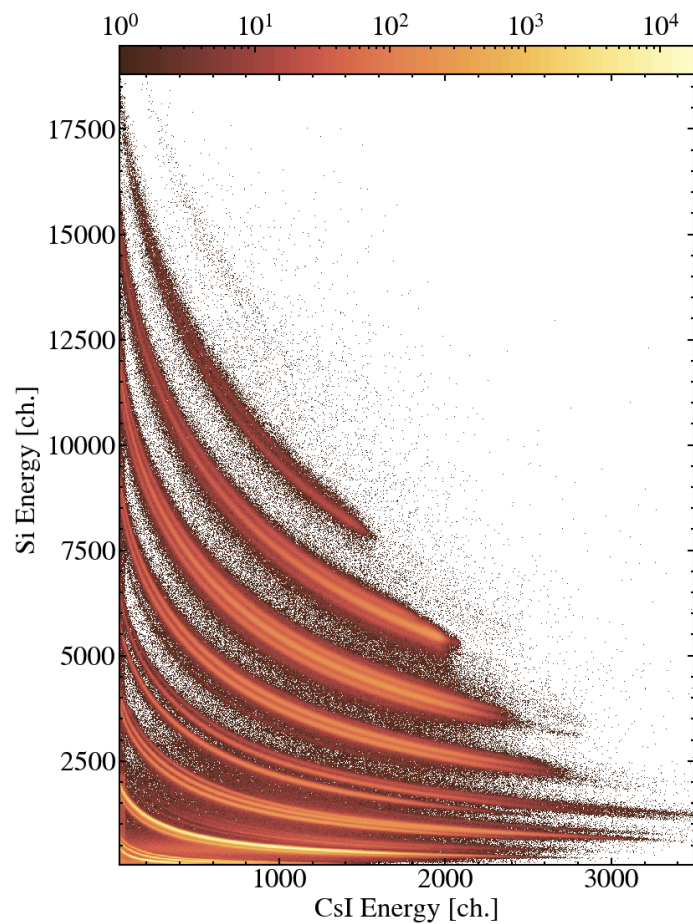
By-hand linearization result



Complete Linearization Method:

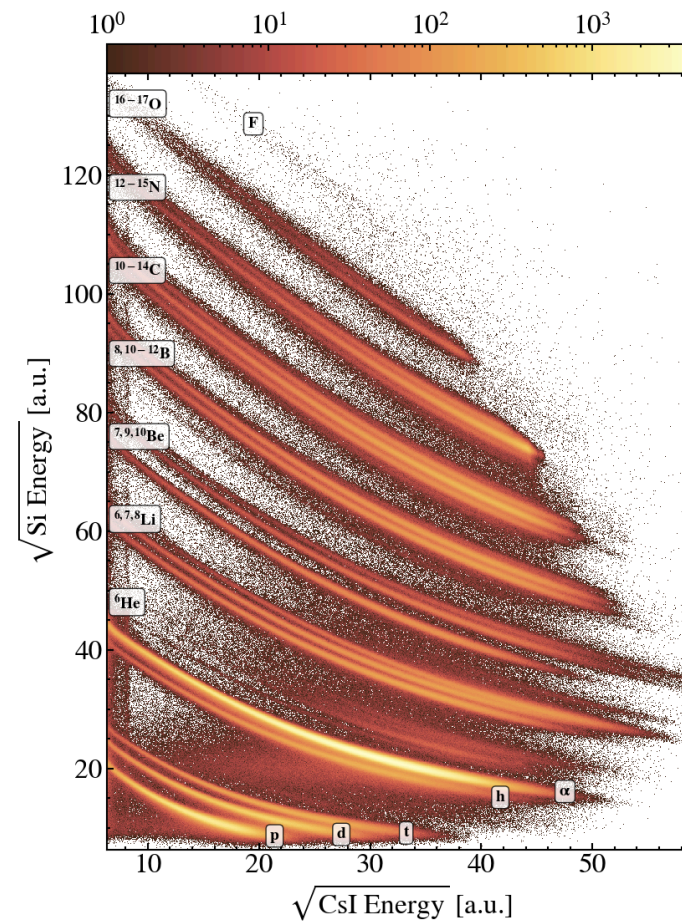
- Find curves which describe the ‘flow’ of data
 - ~~by hand point picking~~
 - Data Equalization
 - Ridge Detection
 - By-hand curve selection
 - Curve Extrapolations
- Straighten the data based on the curves
 - ~~use labelled curves to guide~~
 - Use the natural spacing of the data

Equalizing Data

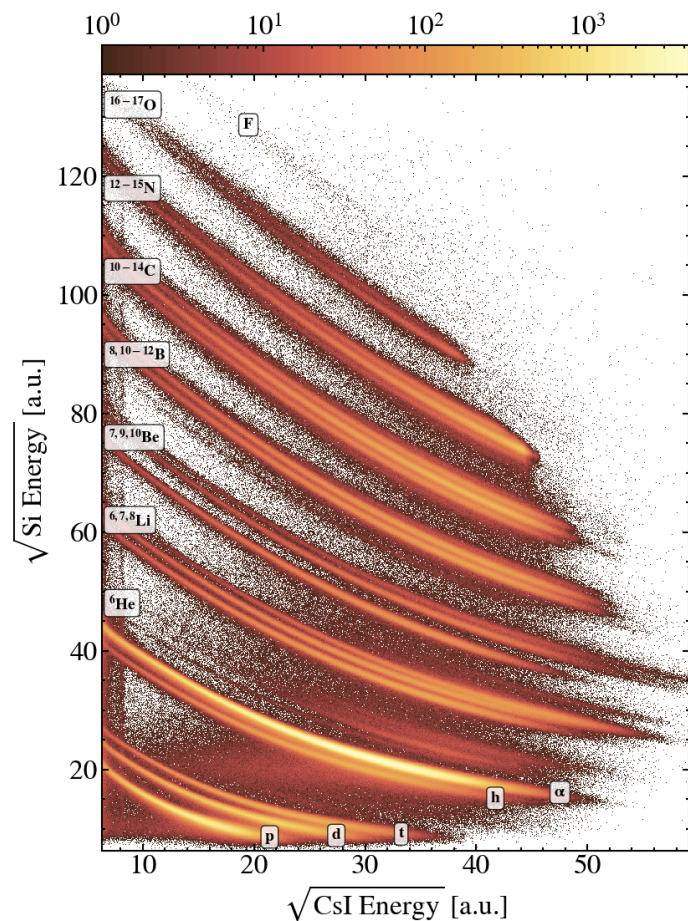


$$-\frac{dE}{dx} \propto Z^2$$

Taking square root of both axes makes the data easier to work with



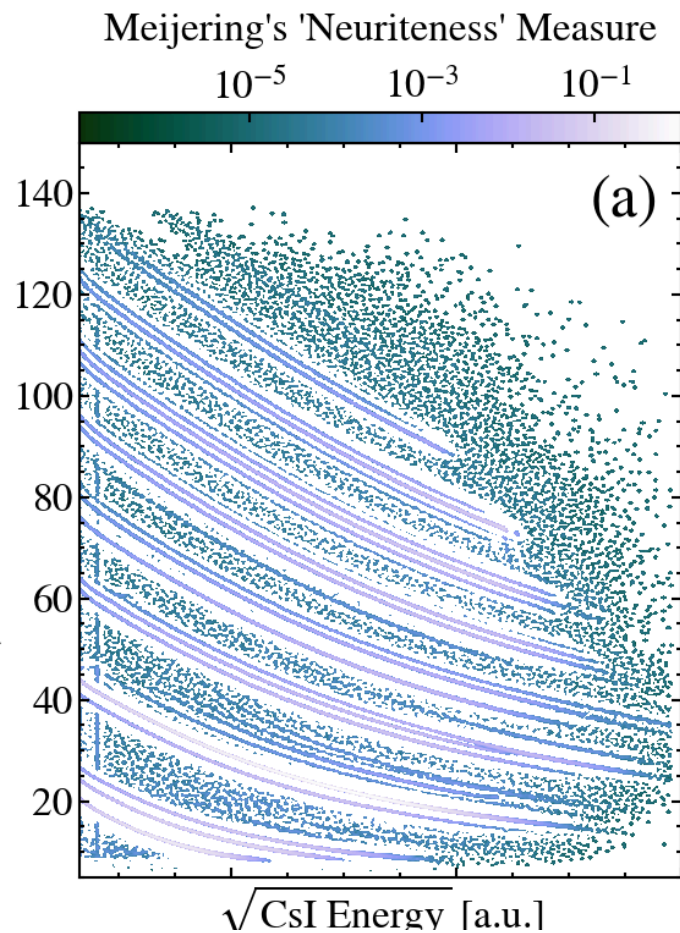
Ridge Detection



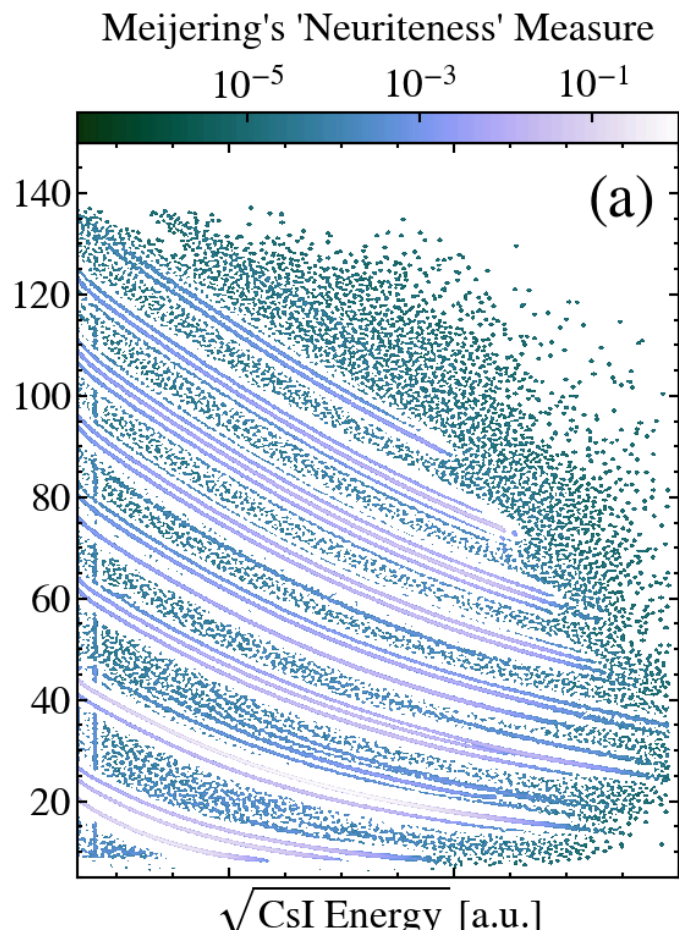
Core idea to replace by-
hand curves

Image based analysis

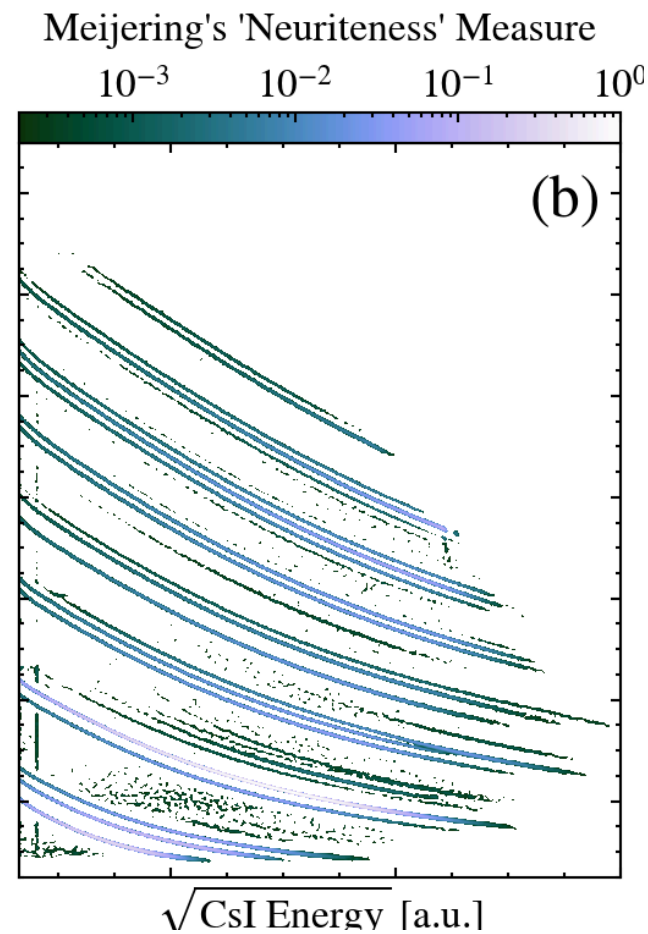
Ridge-Like
=
Pixels which are maxima in
exactly one direction



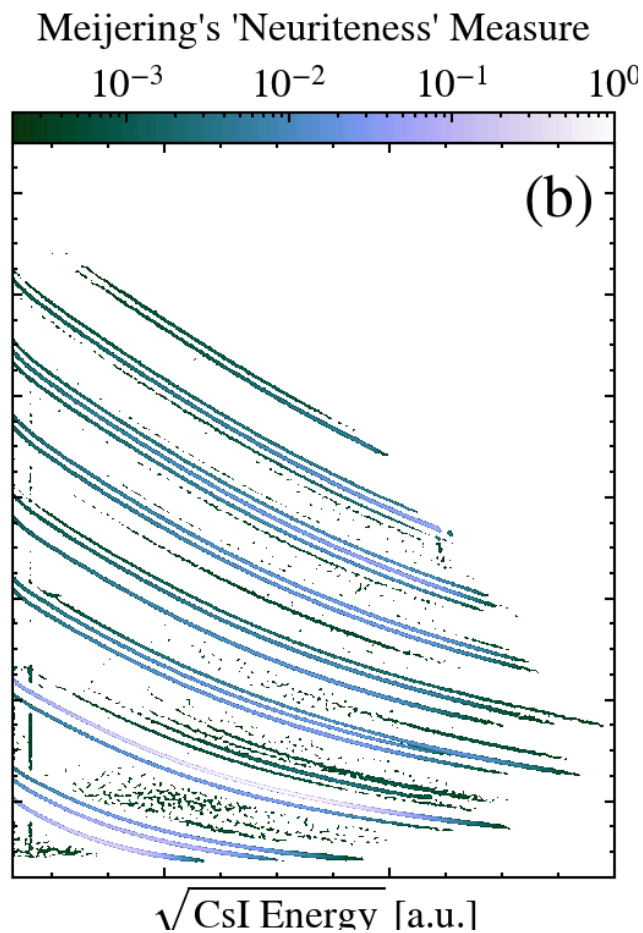
Apply Noise Threshold



First step in isolating
describing curves



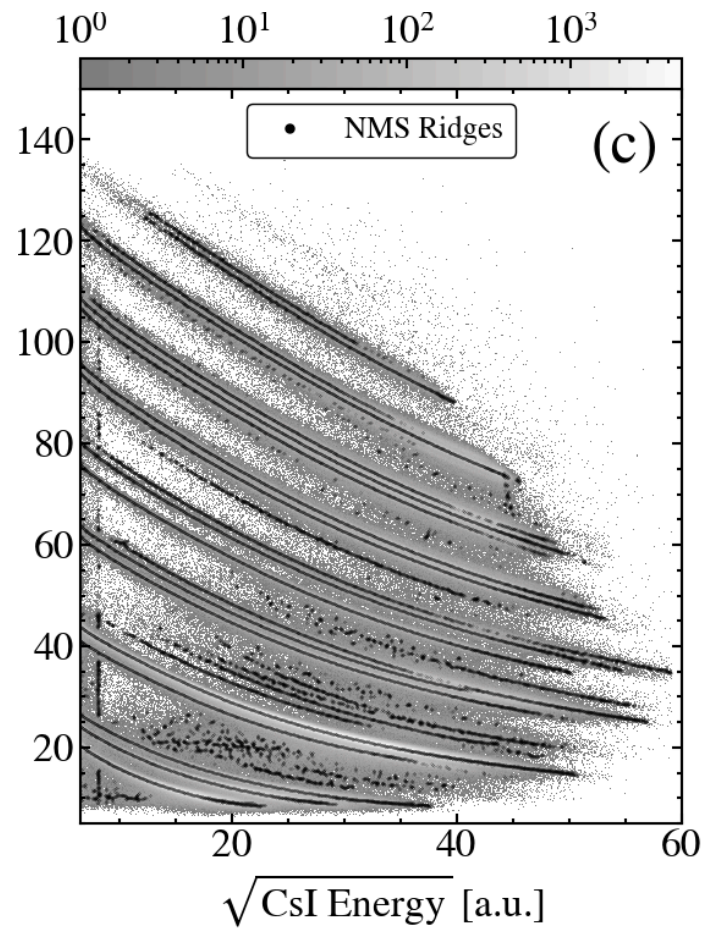
Non-Maximum Suppression



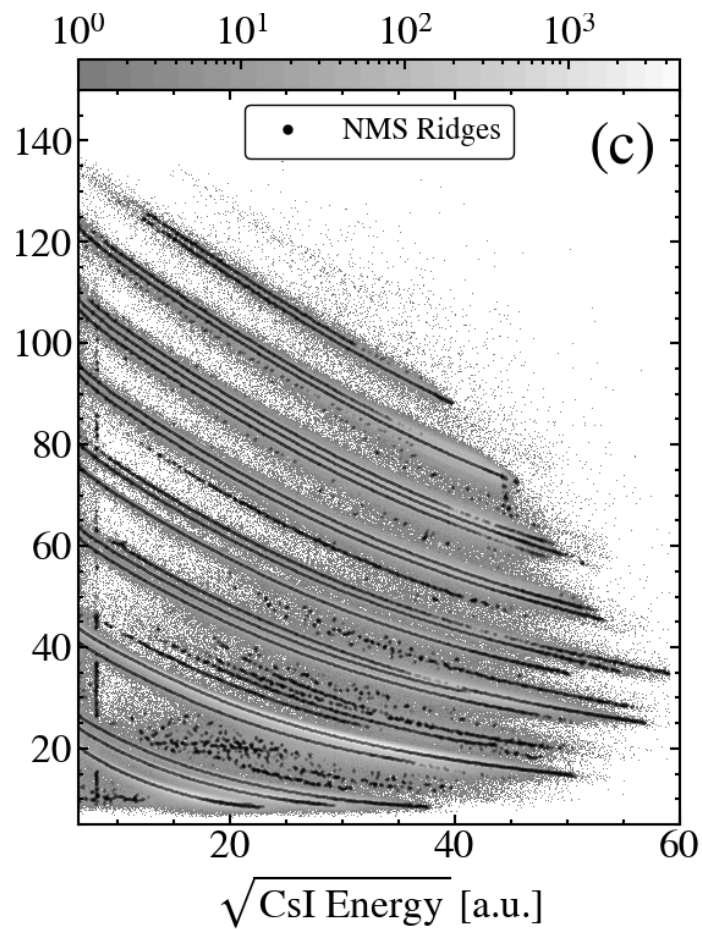
If a pixel is the maximum of
its neighbors in any
direction, keep it
Otherwise, drop it

Probably a
diagram here

Has the effect of narrowing
the curves



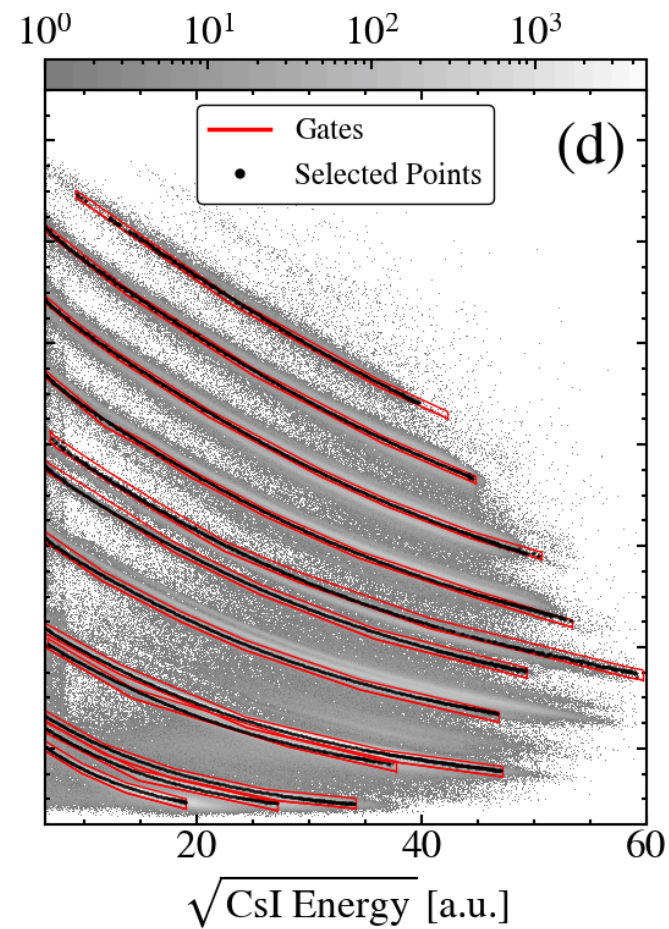
Gate on Ridge-Points



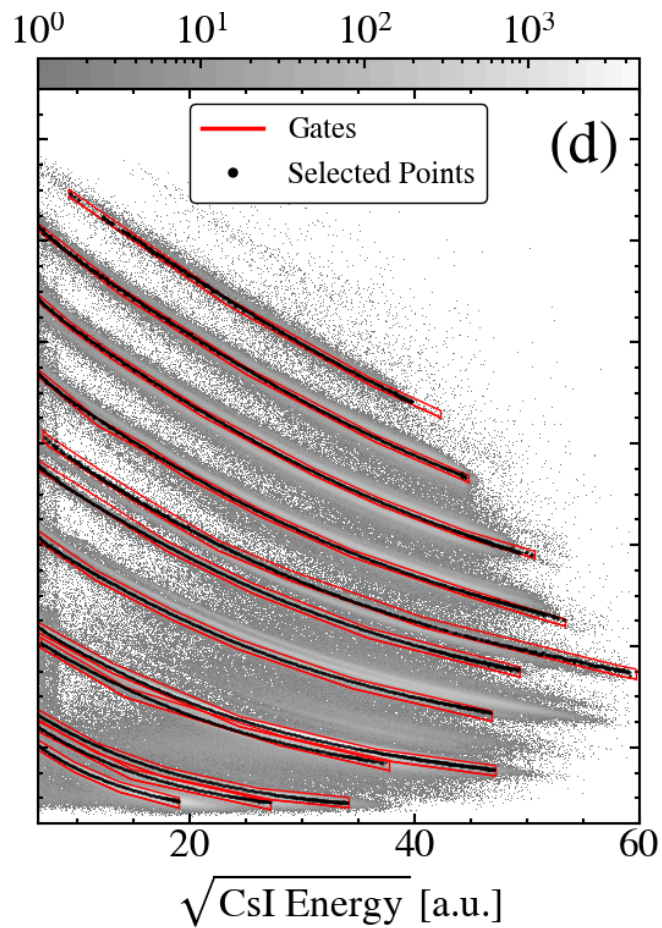
Only by-hand step

Similar in essence to older methods, but MUCH faster

Required precision is much lower

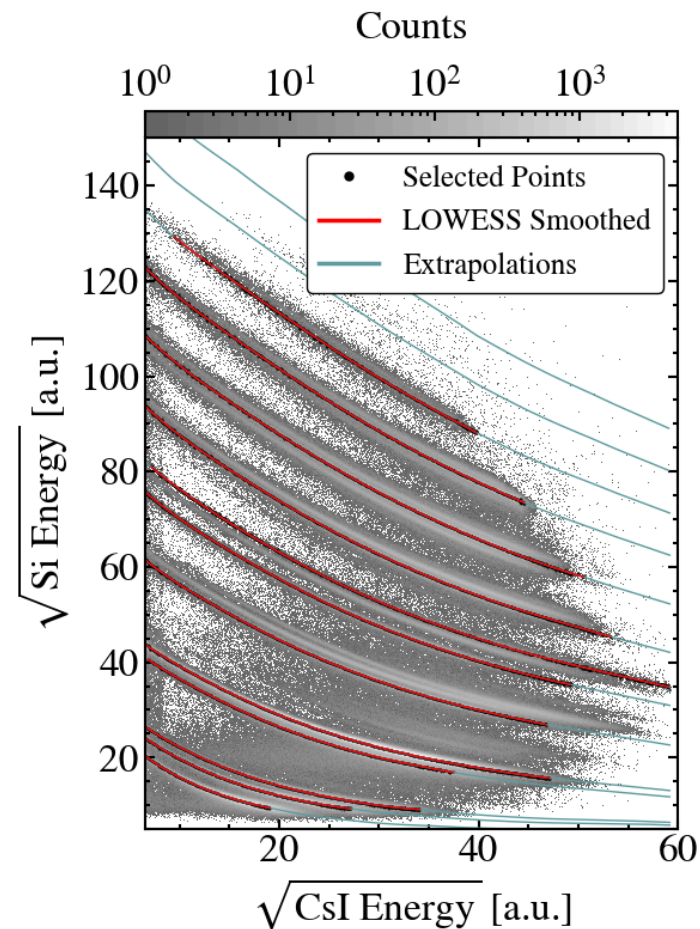


Extrapolate to Edges of Data



Curves cannot be allowed
to cross

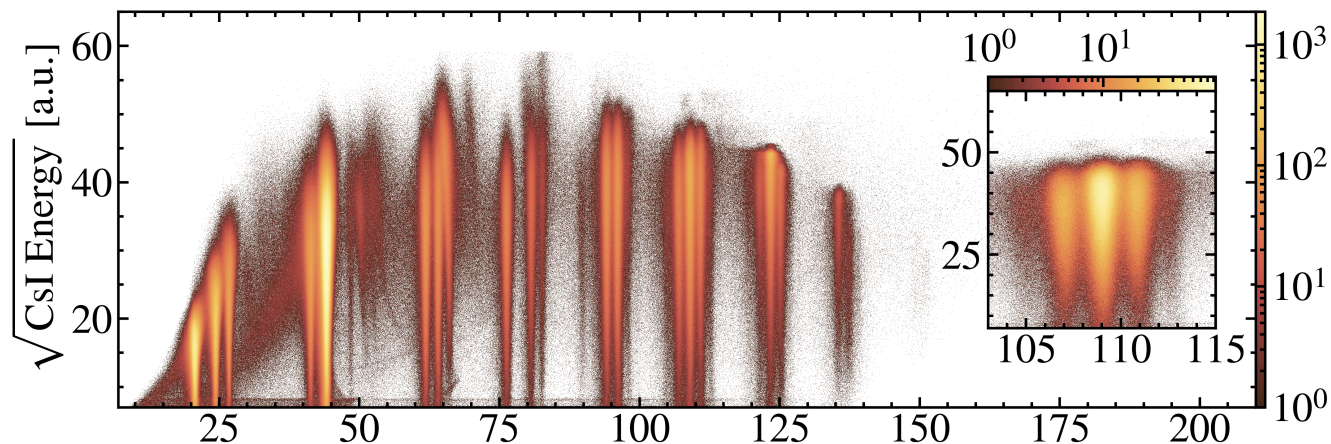
Curves get extrapolated
based off of their curvature
and their neighbors'
curvatures



Linearized Result

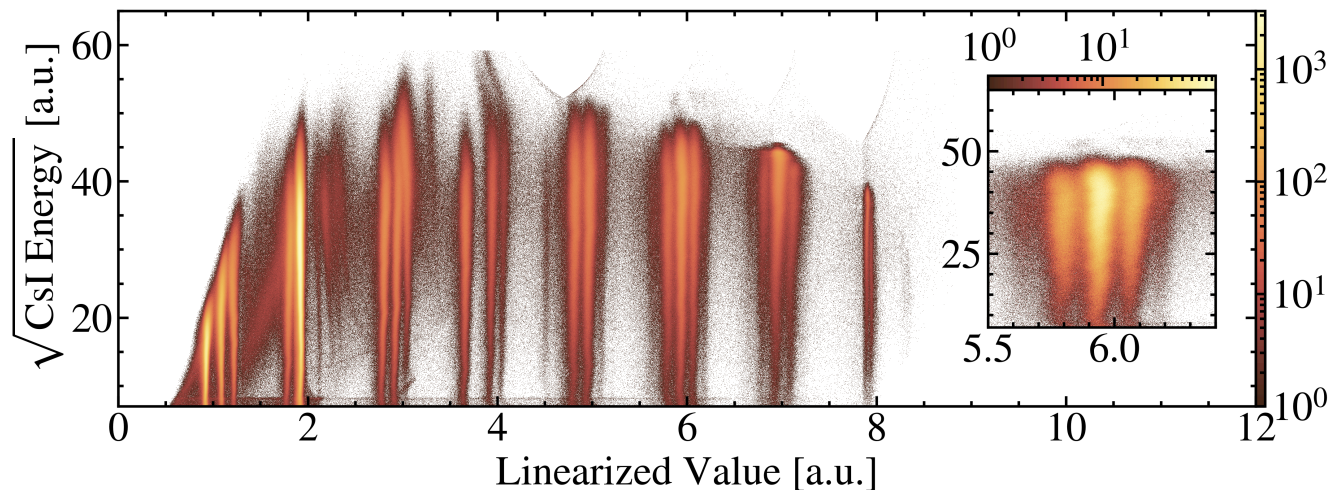
TAPIR Linearization

- ~ **6 minutes** per detector
- Very **linear** bands

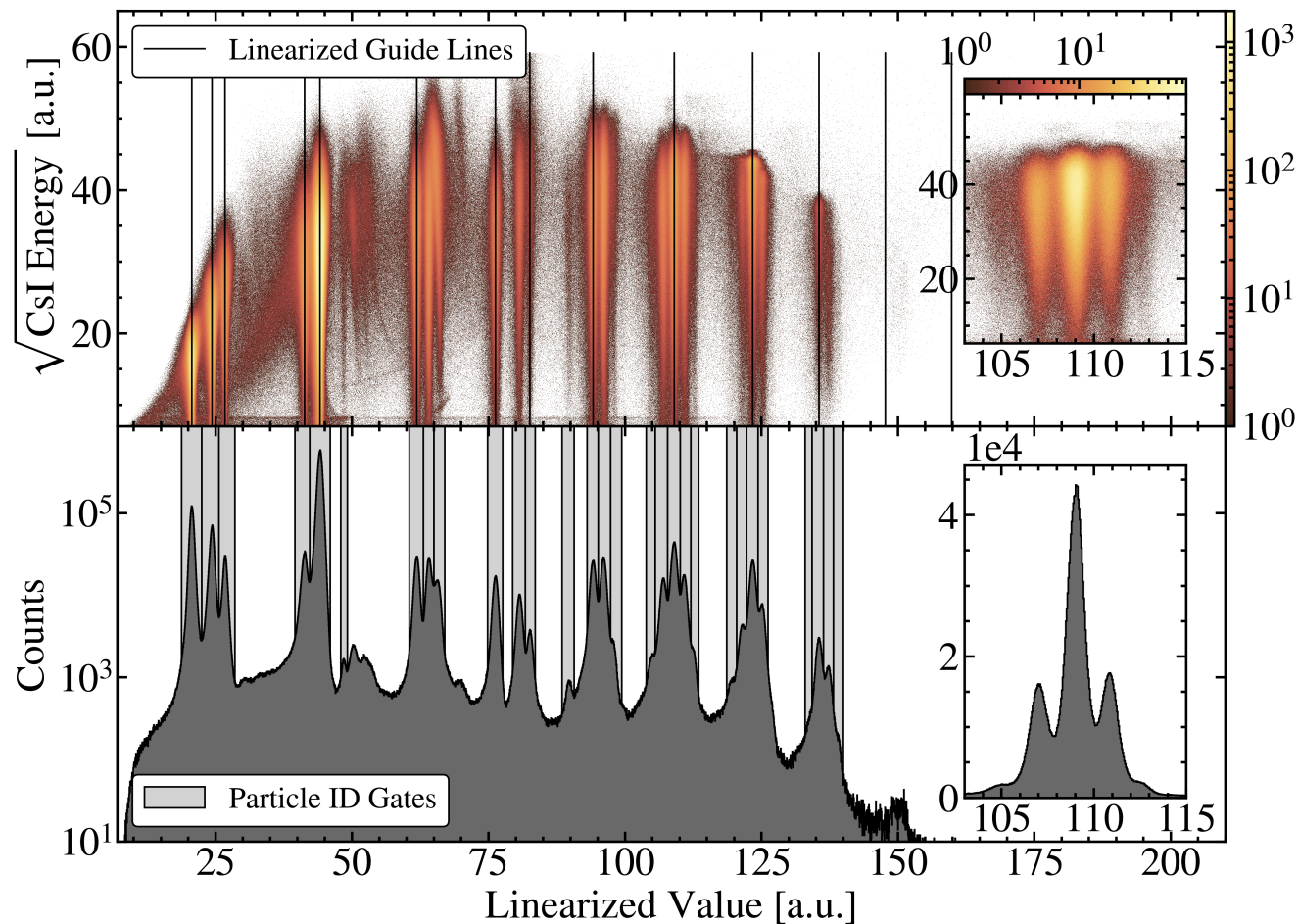
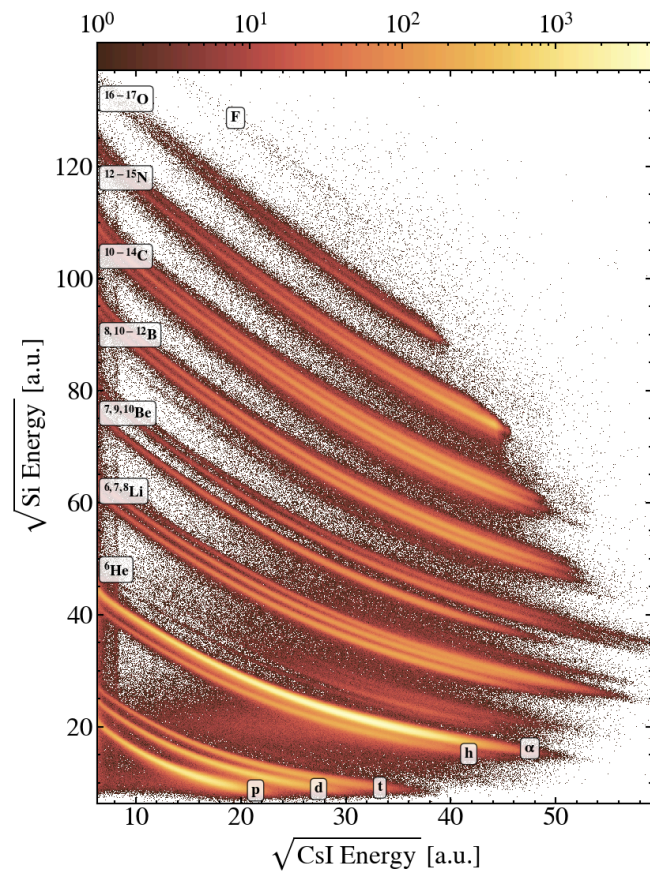


By-Hand Linearization

- Limited to **45 minutes**
- **Wiggly** bands
 - Especially at low E
- Even more time would need to be invested in production



Linearized Result



Acknowledgments and Questions Slides